Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/24/2011
PI Name:	Li, Chuan-Yuan Ph.D.		
Project Title:	A mechanistic investigation of space radiation-induced carcinogenesis		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27710	**Congressional District:**	4
Comments:	PI moved to Duke University in December 2011. Formerly at University of Colorado Denver (Ed., 2/8/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	06/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2012 per NSSC/A. Chu/ARC (Ed., 3/23/2011) NOTE: Received NCE until 5/31/2011 (previous end date 5/31/2010), per A. Chu/ARC (6/2010)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Li, Fang (University of Colorado Denver)		
Grant/Contract No.:	NNX09AH19G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of major concerns for manned space missions of NASA is exposure to galactic cosmic rays (GCRs) or highly charged energetic (HZE) particles, which carries distinct health risks. The major goal of the NASA Bioastroanautics Roadmap and NASA ground-based studies in radiation biology is to assess potential risks of human exposure to HZE particles and to generate knowledge that can be used to mitigate the health risks of HZE particle exposure eventually. In this project, we will two specific aims to study HZE particle-induced mutagenesis and carcinogenesis in mammalian cells. These are: 1. To determine the potential interactions of reactive oxygen/nitrogen species and apoptosis in regulating HZE radiation-induced mutagenesis in mammalian cells. 2) to determine the roles of program cell death in HZE radiation induced mutagenesis/carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research will have the following potential benefit for life on earth: 1) we will develop a state-of-art techniques to monitor radiation induced DNA damage, which will facility a better understanding radiation induced carcinogenesis in humans. 2) Our study may lead to fundamental insights into how cells deal with DNA damage. 3) we hope to achieve novel, mechanistic insights into the carcinogenic risks of radiation, which is universally present on earth.
Task Progress:	In the past year, our efforts have been focused on the following two goals: 1. To develop effective non-invasive, imaging tools to monitor space radiation induced DNA damage and cell death; 2. To use the developed tools to understand molecular mechanisms involved in human cellular response to space radiation and other ionizing radiation. We have achieved substantial progress towards the above goals: 1. We have developed two novel reporters that allow us to monitor space radiation induced DNA damage as well as cell death. 1a. For monitoring DNA damage, we developed a novel, non-invasive way to monitor H2AX foci formation through the so called “split-luciferase” approach. In this approach, firefly luciferase was split into N-terminal and C-terminal halves. These two halves were then fused to H2AX and MDC1 genes, respectively. When there is a DNA double strand break, H2AX would be phosphorylated and associate with MDC1, which reconstitutes luciferase. Therefore, the amount of DNA dsbs could be quantified by measuring the amount of firefly luciferase generated in irradiated cells. Our method was recently published in Cancer Research (W. Li et al. 2011, 71:4130-4137). Using this method, we discovered that Fe ions generated a “second wave” of DNA damage that is not all explainable through apoptosis. We are currently actively studying the mechanism for this interesting and potentially important phenomenon. 1b. Another tool we developed is to monitor cellular caspase activation, which is traditionally used as a bench mark for cellular death. In order to monitor caspase activities in the iPSC induction process, we developed a non-invasive caspase reporters to monitor caspase activities non-invasively. Our caspase reporters consist of a firefly luciferase-GFP fusion protein (LucGFP) linked to a polyubiquitin domain. In between the two moieties, a caspase cleavage site (for example -IETD- for caspase 8 or –DEVD- for caspase 3) is inserted. The assumption is that when caspases are inactive, the reporter proteins will be recognized by proteasomes and degraded immediately because the polyubiquitin domain serves as tag for protein destruction by proteasomes. However, when caspase 3 or 8 is active, the polyubiquitin domain will be cleaved off the reporter protein, leading to enhanced GFP and luciferase signals because these reporter proteins are now not subject to direct proteasome recognition and degradation. Our data indicate that our caspase reporters are powerful and sensitive reporters for caspase activation in vitro in cell culture and in vivo in animal models. Our reporter was published in a recent high impact article (Li et al, Cell Stem Cell, 2011, 7:508-520). 2. We have also studied the roles of apoptotic caspases three diverse biological processes. 2a. We identified a novel, unexpected role for apoptotic cell death in tissue regeneration. We found that dying cells are a potent source of signals for wound healing and tissue regeneration. Most surprisingly, caspases, which are generally thought of as “executioners” in the final stages of apoptosis, were found to play major role in tissue regeneration. We named this newly discovered pathway the “Phoenix Rising” pathway and it was published as a cover story in a high impact journal (Li, et al. 2010, Science Signaling, 3(110); Ra13). [Ed. Note 6/27/2011: also reported in FY2010 report; see Cumulative Bibliography.] 2b. We also identified key roles of apoptotic caspases 3&8 in cellular de-differentiation. Apoptotic caspases facilitate cellular reprogramming process through cleavage of the retinoblastoma protein (Rb). This process could have important implications in radiation induced carcinogenesis, a possibility that is actively investigated in our laboratory currently. Our discovery made the cover story in the journal Cell Stem Cell (Li et al, Cell Stem Cell, 2011, 7:508-520). 2c. Furthermore, we solved a long-standing mystery in radiation biology/oncology: accelerated repopulation of tumor tissues after they were effectively killed by radiation. We found that the “Phoenix rising” pathway, which was described for normal tissue injury, is responsible for tumor repopulation after radiation therapy. Our discovery will be published by the high impact journal Nature Medicine in July, 2011. In summary, we have developed new tools to study space radiation induced DNA damage and cell death and make important discoveries on the roles of caspases in fundamental biological processes such as tissue regeneration, stem cell biology, as well as cancer radiotherapy.
Bibliography Type:	Description: (Last Updated: 10/30/2019)
Articles in Peer-reviewed Journals	Li F, He Z, Shen J, Huang Q, Li W, Liu X, He Y, Wolf F, Li C-Y. "Apoptotic caspases regulate induction of iPSCs from human fibroblasts." Cell Stem Cell. 2010 Oct 8;7(4):508-20. PMID: 20887956 , Oct-2010
Articles in Peer-reviewed Journals	Li W, Li F, Huang Q, Shen J, Wolf F, He Y, Liu X, Hu YA, Bedford JS, Li C-Y. "Quantitative, noninvasive imaging of radiation-induced DNA double-strand breaks in vivo." Cancer Research 2011 Jun 15;71(12):4130-7. Epub 2011 Apr 28. PMID: 21527553 , Jun-2011

Articles in Peer-reviewed Journals	Wolf F, Li W, Li F, Li C-Y. "Novel luciferase-based reporter system to monitor activation of ErbB2/Her2/neu pathway noninvasively during radiotherapy." Int J Radiat Oncol Biol Phys. 2011 Jan 1;79(1):233-8. Epub 2010 Oct 8. PMID: 20934271 , Jan-2011
Articles in Peer-reviewed Journals	Huang Q, Li F, Liu X, Li W, Shi W, Liu F-F, O'Sullivan B, He Z, Peng Y, Tan A-C, Zhou L, Shen J, Han J, Wang X-J, Thorburn J, Thorburn A, Jimeno A, Raben D, Bedford JS, Li C-Y. "Caspase 3-mediated stimulation of tumor cell repopulation during cancer radiotherapy." Nature Medicine, in press, June 2011. , Jun-2011
Journal/Magazine covers	Li F, He Z, Shen J, Huang Q, Li W, Liu X, He Y, Wolf F, Li CY. "Cover in journal Cell Stem Cell for article, 'Apoptotic caspases regulate induction of iPSCs from human fibroblasts.' " Cell Stem Cell. 2010 Oct 8;7(4):508-20. http://www.cell.com/cell-stem-cell/issue?pii=S1934-5909%2810%29X0010-8 , Oct-2010